WASHAKH ALI

AI / ML Engineer | LLMs, RAG & Agentic Systems

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SUMMARY

AI/ML Engineer, 3+ years shipping production GenAI, from diffusion APIs serving 300K+ generations to production RAG and function-calling assistants grounded in live company content and databases. Backend-strong: FastAPI, async Python, streaming, Docker, and CI/CD, with a track record of taking systems from prototype to secure, deployed production.

SELECTED PROJECTS

- **Graxo Cybersecurity RAG Chatbot** [Live]: Designed and built a production RAG chatbot from scratch for a cybersecurity consultancy, replacing an ungrounded LLM-passthrough system with one answering strictly from real company content. Structure-aware content ingestion pipeline generating Gemini embeddings indexed in Pinecone, and a clean OO FastAPI backend. Also audited and hardened the prior system's live security gaps.
- **SiteSpend — LLM Expense Assistant** [Code]: Natural-language chat over a live production database using Gemini function-calling (6 tools) with token-by-token SSE streaming; deployed as an independent serverless service with zero deploy coupling to the host app.
- **AI-Powered Job-Search Automation** [Code] : Full-stack async app with a plugin-based scraper registry across multiple job sources and LLM-based candidate-job matching using a structured scoring rubric.Docker/Compose deployment with production hardening owned solo end to end and open-sourced (MIT).

CORE SKILLS

LLMs, RAG & Agentic AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Vector Databases (Chroma, Pinecone), Embeddings & Semantic Search, Asymmetric Retrieval (query/document modes), Tool-Calling / Function-Calling, Multi-Turn Memory, Agentic AI, Model Context Protocol (MCP), LangChain, LlamaIndex, Transformers, LLM Fine-Tuning, Prompt Engineering & Prompt-Injection Defense, Gemini API, , Ollama (self-hosted LLMs), OpenAI API, Hugging Face, n8n,

Data & Ingestion: : Content Ingestion Pipelines, Web Scraping (Playwright, Apify), Plugin-Based Scraper Architecture, DOM-Aware Parsing & Extraction, Heading-Aware Chunking, Embedding Generation, Cross-Source Deduplication

Backend & Deployment: Python, FastAPI, Pydantic (v2), asyncio, async SQLAlchemy 2.0 (ORM), Alembic Migrations, Dependency Injection, Server-Sent Events (SSE) / Streaming, REST APIs, APScheduler (Background Jobs & Scheduling), Docker (multi-stage builds), Serverless (Vercel), Supabase / PostgreSQL, Firebase, uv, Caching, Rate-Limit & Error Handling, GitHub Actions (CI/CD), Ruff, Pyright

Auth, Security & Testing: JWT / OAuth2 Password Flow, Argon2 Password Hashing, Ownership-Scoped Authorization, API Security (CORS, key-exposure remediation, input validation), Security Headers (HSTS, CSP), pytest / pytest-asyncio, httpx Test Client, Systematic Evidence-Based Debugging, Secrets / PII Hygiene

ML & Computer Vision: PyTorch, TensorFlow, Keras, Diffusers, Stable Diffusion, CNNs, Multi-Head Detection, OpenCV, PaddleOCR, scikit-learn, NumPy, Pandas, Time-Series Forecasting

Inference & Systems: CUDA, Multi-GPU Inference, GPU Memory Management, Parallel Inference Pipelines, Model Layer Caching

Integrations & Frontend: Third-Party API Integration (RapidAPI, ATS boards), Google Sheets API, Cross-Source Deduplication, Jinja2, Vanilla JS (fetch, async/await), Bootstrap 5, Responsive CSS

PROFESSIONAL EXPERIENCE

ML Engineer — Runware (Contractor) Jan 2025 – Aug 2026

Production Image & Video Generation API

- 300,000+ production image/video generations sustained across concurrent GPU models, engineered as diffusion-based generation APIs in FastAPI with a custom GPU memory-management system that held under high-concurrency load.
- Reduced inference latency on concurrent production workloads by architecting parallel multi-GPU inference pipelines and optimizing model layer caching, without degrading output quality.

LLM & Model Integration

- Integrated the latest open-source models across image, video, audio, LLM and 3D modalities into a unified FastAPI inference layer, expanding platform capabilities well beyond 2D generation.
- Cut compute cost per generation by optimizing these models for faster inference with CUDA parallel computing, multi-GPU execution and layer caching, so the company served more workloads on the same GPU fleet.
- Enabled reproducible, portable inference across environments by containerizing model services with Docker.

Computer Vision Engineer — Vision RD Dec 2023 – Dec 2024

Diffusion API & Automated Inspection

- 98.2% defect-detection accuracy across 4 defect classes (polarity errors, missing/shifted components, solder issues), achieved by designing a multi-head deep learning model for automated PCB inspection and fine-tuning PaddleOCR for component name/value extraction.
- Reduced inference latency and GPU memory footprint on a Stable Diffusion image-generation API by researching and implementing multi-GPU inference and applying published model-layer-caching techniques.

Data Analyst — Central Power Purchasing Agency (CPPA-G) Jul 2023 – Dec 2023

- Improved national power-consumption planning by building predictive time-series forecasting models on historical grid data.
- Increased stakeholder interpretability of technical reports by applying generative AI to auto-produce narrative data summaries.

EDUCATION

Bachelor of Computer Engineering — Air University, Islamabad Sep 2019 - Jun 2023

CERTIFICATIONS

- Agentic AI: RAG & MCP Chatbots (Coursera)
- Generative AI with Diffusion Models — NVIDIA
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI
- Advanced Computer Vision with TensorFlow — DeepLearning.AI
- Getting Started with TensorFlow 2 — Imperial College London
- Artificial Intelligence Workshop — Dice Analytics
- Fundamentals of Digital Image and Video Processing — Northwestern University
- Python Data Structures — University of Michigan